G.1.1.4 Ito et al. (2013)

Ito et al. (2013) used factor analysis to characterize pollution sources and assess the association between short-term $PM_{2.5}$ and $PM_{2.5}$ components with morbidity and mortality outcomes. The authors performed a city-level time series analysis using mortality data from the National Center for Health Statistics (NCHS). They included 150 U.S. cities (and 64 cities in a second analysis) in their assessment of 24-hour all-cause mortality for all ages. $PM_{2.5}$ exposure estimates were based on average daily monitored $PM_{2.5}$ mass data in each city. $PM_{2.5}$ concentrations were expressed in the model as a deviation from the monthly mean to reduce the influence of the seasonal cycles of the pollutants on the overall associations and help focus on the short-term associations. The authors first ran city- and season-specific Poisson generalized linear models and then combined the city-specific estimates using a random effects approach. The model adjusted for temporal trends (annual cycles and influenza epidemics), immediate and delayed temperature, and day-of-week pattern, for entire years (2001-2006) and for warm (April-September) and cold (October-March) seasons. The authors also assessed effect modification using land-use variables and average air pollution levels.

Short-term Mortality, All Cause

Ito et al. (2013) provided regression coefficients and standard errors which are directly used in BenMAP-CE. The selected effect estimate is the lag-1 day (all year) beta of 0.000145 (0.0000747) (Ito, et al., 2013, Appendix G, Table G.6 for Figure 4).

G.1.1.5 Jerrett et al. (2017)

Jerrett et al. (2017) compared mortality effect estimates for $PM_{2.5}$ modeled from remote sensing to those for $PM_{2.5}$ modeled using ground-level information. Multiple exposure estimation approaches were evaluated and the risk assessment used results based on an ensemble approach that incorporated chemical transport modeling, land use data, satellite data, and data from ground-based monitors. This cohort study evaluated long-term $PM_{2.5}$ exposure and mortality from IHD and diseases of the circulatory system in the ACS Cancer Prevention Study II cohort (ages 30+). The authors utilized a Cox proportional hazards model controlling for individual-level confounders which included current and former smoking status as well as smoking duration, amount, age started, second hand cigarette smoke (hours/day exposed), exposure to $PM_{2.5}$ in the workplace for each of the subject's major lifetime occupation, self-reported exposure to dust/fumes at work, marital status, level of education, BMI, alcohol consumption, dietary vegetable/fruit/fiber index, dietary fat index, and missing nutrition information. Ecologic characteristics included median household income, percentage of people with <125% of poverty-level income, percentage of persons >16 who are unemployed, percentage of adults with <12th grade education, and percentage of population who were Black or Hispanic.

Mortality, Ischemic Heart Disease (ICD-10 code I20-I25)

In a single-pollutant model, the coefficient and standard error are estimated from the hazard ratio (1.15) and 95% confidence intervals (95% CI: 1.11-1.19) for a 10 μg/m^3 increase in the ensemble estimate of $PM_{2.5}$ exposure level from 2002-2004 (Jerrett, et al., 2017, Table 4 IHD). The model was fully adjusted for ecological confounders.

G.1.1.6 Krewski et al. (2009)

This cohort study consists of approximately 360,000 participants residing in areas of the country that have adequate monitoring information on levels of $PM_{2.5}$ for 1980 and about 500,000 participants in areas with adequate information for 2000. The causes of death that were analyzed included all causes, cardiopulmonary disease (CPD), ischemic heart disease (IHD), lung cancer, and all remaining causes. Data for 44 personal, individual-level covariates, based on participants' answers to a 1982 enrollment questionnaire, were also used for the analyses. The authors also collected data for seven ecologic (neighborhood-level) covariates, each of which represents local factors known or suspected to influence mortality, such as poverty level, level of education, and unemployment (at both Zip Code and city levels). Long-term average exposure variables were constructed for $PM_{2.5}$ from monitoring data for two periods: 1979-1983 and 1999-2000. Similar variables were constructed for long-term exposure to other pollutants of interest from single-year (1980) averages, including total suspended particles, ozone, nitrogen dioxide, and sulfur dioxide. Exposure was averaged for all monitors within a metropolitan statistical area (MSA) and assigned to participants according to their Zip Code area (ZCA) of residence. The authors chose the standard Cox proportional-hazards model (and a variation to allow for random effects) to calculate hazard ratios for various cause-of-death categories associated with the levels of air pollution exposure in the cohort. They extended the random effects Cox model to accommodate two levels of information for clustering and for ecologic covariates. Three main analyses were conducted: a Nationwide Analysis, Intra-Urban Analyses in the New York City (NYC) and Los Angeles (LA) regions, and an analysis designed to investigate whether critical time windows of exposure to pollutants might have affected mortality in the cohort.

Mortality, All-Cause

In a random effects Cox model, the coefficient and standard error in BenMAP-CE are estimated from the relative risks (1.06) and 95% confidence intervals (95% CI: 1.04-1.08) for a 10 μg/m^3 increase in the average of $PM_{2.5}$ exposure level for 1999-2000 (Krewski, et al., 2009, Commentary Table 4). The results were adjusted for the 44 individual-level covariates and the 7 ecologic covariates at the MSA & DIFF levels.

Mortality, Ischemic Heart Disease (ICD-10 code I20-I25)

In a random effects Cox model, the coefficient and standard error are estimated from the relative risks (1.24) and 95% confidence intervals (95% CI: 1.19-1.29) for a 10 μg/m^3 increase in the average of $PM_{2.5}$ exposure level for 1999-2000 (Krewski, et al., 2009, Commentary Table 4). The results were adjusted for the 44 individual-level covariates and the 7 ecologic covariates at the MSA & DIFF levels.

Mortality, Lung Cancer (ICD-10 code C30-C39)

In a random effects Cox model, the coefficient and standard error are estimated from the relative risks (1.14) and 95% confidence intervals (95% CI: 1.06-1.23) for a 10 μg/m^3 increase in the average of $PM_{2.5}$ exposure level for 1999-2000 (Krewski, et al., 2009, Commentary Table 4). The results were adjusted for the 44 individual-level covariates and the 7 ecologic covariates at the MSA & DIFF levels.

G.1.1.7 Laden et al. (2006)

A large body of epidemiologic literature has found an association of increased fine particulate air pollution ($PM_{2.5}$) with acute and chronic mortality. The effect of improvements in particle exposure is less clear. Earlier analysis of the Harvard Six Cities adult cohort study showed an association between long-term ambient $PM_{2.5}$ and mortality between enrollment in the mid-1970's and follow-up until 1990. The authors extended mortality follow-up for eight years in a period of reduced air pollution concentrations. Annual city-specific $PM_{2.5}$ concentrations were measured between 1979-1988, and estimated for later years from publicly available data. E xposure was defined as (1) city-specific mean $PM_{2.5}$ during the two follow-up periods, (2) mean $PM_{2.5}$ in the first period and change between these periods, (3) overall mean $PM_{2.5}$ across the entire follow-up, and (4) year-specific mean $PM_{2.5}$. Mortality rate ratios were estimated with Cox proportional hazards regression controlling for individual risk factors. The authors found an increase in overall mortality associated with each 10 μg/m^3 increase in $PM_{2.5}$ modeled either as the overall mean (RR=1.16, 95%CI=1.07-1.26) or as exposure in the year of death (RR=1.14, 95%CI=1.06-1.22). $PM_{2.5}$ exposure was associated with lung cancer (RR=1.27, 95%CI=0.96-1.69) and cardiovascular deaths (RR=1.28, 95%CI=1.13-1.44). Improved overall mortality was associated with decreased mean $PM_{2.5}$ (10 microg/m(3)) between periods (RR=0.73, 95% CI=0.57-0.95). Total, cardiovascular, and lung cancer mortality were each positively associated with ambient $PM_{2.5}$ concentrations. Reduced $PM_{2.5}$ concentrations were associated with reduced mortality risk.

All-Cause Mortality

The coefficient and standard error for $PM_{2.5}$ are estimated from the relative risk (1.16) and 95% confidence interval of (1.07-1.26) associated with a change in annual mean exposure of 10.0 μg/m3 (Laden et al., 2006, p. 667).

G.1.1.8 Lepeule et al. (2012)

Lepeule et al. (2012) evaluated the sensitivity of previous Six Cities results to model specifications, lower exposures, and averaging time using eleven additional years of cohort follow-up that incorporated recent lower exposures. The authors found significant associations between $PM_{2.5}$ exposure and increased risk of all-cause, cardiovascular and lung cancer mortality. The authors also concluded that the concentration-response relationship was linear down to $PM_{2.5}$ concentrations of 8 μg/m3, and that mortality rate ratios for $PM_{2.5}$ fluctuated over time, but without clear trends, despite a substantial drop in the sulfate fraction.

G.1.1.9 Pope et al. (2002)

The Pope et al. (2002) analysis is a longitudinal cohort tracking study that uses the same American Cancer Society cohort as the original Pope et al. (1995) study, and the Krewski et al. (2000) reanalysis. Pope et al. (2002) analyzed survival data for the cohort from 1982 through 1998, 9 years longer than the original Pope study. Pope et al. (2002) also obtained $PM_{2.5}$ data in 116 metropolitan areas collected in 1999, and the first three quarters of 2000. This is more metropolitan areas with $PM_{2.5}$ data than was available in the Krewski reanalysis (61 areas), or the original Pope study (50 areas), providing a larger size cohort.

They used a Cox proportional hazard model to estimate the impact of long-term PM exposure using three alternative measures of $PM_{2.5}$ exposure; metropolitan area-wide annual mean PM levels from the beginning of tracking period (1979-1983 PM data, conducted for 61 metropolitan areas with 359,000 individuals), annual mean PM from the end of the tracking period (1999-2000, for 116 areas with 500,000 individuals), and the average annual mean PM levels of the two periods (for 51 metropolitan areas, with 319,000 individuals). PM levels were lower in 1999-2000 than in 1979-1983 in most cities, with the largest improvements occurring in cities with the highest original levels.

Pope et al. (2002) followed Krewski et al. (2000) and Pope et al. (1995, Table 2) and reported results for all-cause deaths, lung cancer (ICD-9 code: 162), cardiopulmonary deaths (ICD-9 codes: 401-440 and 460-519), and “all other” deaths. All-cause mortality includes accidents, suicides, homicides and legal interventions. The category “all other” deaths is all-cause mortality less lung cancer and cardiopulmonary deaths. Like the earlier studies, Pope et al. (2002) found that mean $PM_{2.5}$ is significantly related to all-cause and cardiopulmonary mortality. In addition, Pope et al. (2002) found a significant relationship with lung cancer mortality, which was not found in the earlier studies. None of the three studies found a significant relationship with “all other” deaths.

Pope et al. (2002) obtained ambient data on gaseous pollutants routinely monitored by EPA during the 1982-1998 observation period, including SO_2, NO_2, CO, and ozone. They did not find significant relationships between NO_2, CO, and ozone and premature mortality, but there were significant relationships between SO_4 (as well as SO_2), and all-cause, cardiopulmonary, lung cancer and “all other” mortality.

All-Cause Mortality, 1979-1983 Exposure

The coefficient and standard error for $PM_{2.5}$ using the 1979-1983 PM data are estimated from the relative risk (1.04) and 95% confidence interval (1.01-1.08) associated with a change in annual mean exposure of 10.0 μg/m^3 (Pope et al., 2002, Table 2).